

HASSAN RAZA

RF & Wireless Engineer · NOC Analyst · Researcher

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PROFILE

Data-driven RF & Wireless Engineer and NOC Analyst with 3+ years of experience across Distributed Antenna Systems (DAS), wireless network optimization and data-driven post-processing. Strong in Python automation, wireless KPIs and drive-test tooling. Passionate about AI/ML for next-generation (B5G & 6G) networks, federated learning and secure communications — seeking research collaboration and international exposure in 5G/6G innovation.

PROFESSIONAL EXPERIENCE

NOC Engineer II

Dec 2024 — Present

DAS Technology Services — Islamabad (Remote)

- Monitor, analyze and resolve DAS network alarms and faults to ensure continuous, optimal uptime.
- Coordinate with field and integration teams to fix remote and on-site issues, minimizing downtime.
- Use Gladiator, Wind Console and Wireshark for real-time monitoring, fault analysis and alarm correlation.
- Generate detailed incident reports and maintain network configuration and troubleshooting documentation.

RF Engineer II

Oct 2023 — 2024

DAS Technology Services — Islamabad

- Designed and optimized iDAS/oDAS networks using iBwave for high-capacity venues such as hospitals and stadiums.
- Conducted drive-test surveys with TEMS, Gladiator and Wind Console, capturing RSRP, RSRQ and SINR.
- Performed RF post-processing, data cleaning and KPI evaluation to detect coverage gaps and interference.
- Produced performance/KPI reports to 3GPP/IEEE standards; trained and mentored junior engineers.

Data Engineer Intern

Feb 2023 — Oct 2023

TMC (Tally Marks Consulting) — Karachi

- Built Python pipelines for data scraping, cleaning and transformation using BeautifulSoup and Selenium.
- Developed data dashboards and supported data governance, security and quality initiatives.

KEY PROJECTS

CryptoSense AI (Python · Flutter · FastAPI · ML) — crypto-forecasting platform; LightGBM/XGBoost/RandomForest models served via FastAPI, with a Flutter app (web + Android) showing live charts, bull/bear signals and a Claude-powered chat assistant.

DAS Fault AI (Python · ML) — AI-driven fault detection for Distributed Antenna Systems, surfacing anomalies and likely root causes from network telemetry.

Auto-Timetable Generator (FastAPI · React/TS · OR-Tools) — conflict-free university timetable solver using Google OR-Tools CP-SAT, resolving real scheduling constraints in under a second.

RF Post-Processing Suite (Python) — KPI summary calculator, CSV collector/customizer and PowerPoint alignment & weighting utilities for RF reporting.

NOC Alarm Analysis Dashboard (Python) — correlates and analyzes NOC network alarms to speed up fault diagnosis and resolution.

More projects at github.com/HassanRaz-A

EDUCATION

BS in Computer Science & IT

2019 — 2023

NED University of Engineering & Technology — Karachi

- Final Year Project: AI-Based Class Scheduling & Classroom Allocation System — reduced scheduling errors by ~95% through algorithmic optimization.

Higher Secondary Certificate (Pre-Engineering)

FBISE — Islamabad

2019

Secondary School Certificate (Science)

FBISE — Islamabad

2016

TECHNICAL SKILLS

Programming: Python (automation, data analysis, scraping), SQL

RF & Wireless: RF Design, DAS (iDAS/oDAS), RF Drive Testing, Post-Processing, KPI Analysis, Network Optimization

Tools: iBwave, TEMS Investigation, Gladiator, Wind Console, Wireshark, BeautifulSoup, Selenium, MATLAB

Data & AI/ML: Data Cleaning, Pipeline Building, Data Visualization, Machine Learning, AI-based fault detection

Soft Skills: Cross-functional Collaboration, Mentoring & Training, Documentation, Troubleshooting

CERTIFICATIONS & TRAINING

- 5G Network Architecture & Key Technologies — Huawei (2024)
- Google Cybersecurity Specialization — Google (2024)
- Fundamentals of Private Networks — iBwave (2024)
- RF Fundamentals for Public Safety — iBwave (2024)
- 4G-LTE Drive Test & Post Processing — Udemy (2024)
- ALLIANCE Multi-Carrier DAS — SOLiD University (2025)
- Wireshark Essential Training — LinkedIn (2025)
- Critical Infrastructure Protection (CIP) — OPSWAT Academy (2025)
- Satellite Internet & Communications — LinkedIn (2024)
- Android Malware Analysis — Udemy (2023)

LANGUAGES

English — Fluent **Urdu** — Native